

Overview of AWS

Amazon Web Services (AWS) is one of the Cloud-based Infrastructure Management System (IMS) solutions available in the market. It is an end-to-end service provider of cloud computing services right from offering compute power, database storage, content delivery, and other functionalities that enable your business to scale and grow exponentially. What began in 2006 as a method of offering IT infrastructure services to companies in the form of web services (now commonly known as cloud computing) has evolved into a highly reliable, scalable, low-cost infrastructure platform with an incredible computing capacity that powers many a business spanning multiple countries and time zones.

Overview of AWS IAM

AWS Identity Access Management (IAM) enables you to control access to your AWS resources safely, securely, and efficiently. IAM is used to create AWS users, groups, roles, and policies.

Key Benefits

- Create users and assign individual security credentials (access keys, passwords, and/or Multi-Factor Authentication devices) or request temporary security credentials to provide access to AWS resources.
- Enable identity federation to allow existing identities from your corporate directory or third-party websites like Amazon, Facebook, Google, etc., to access the AWS Management Console, call AWS APIs, and access resources without creating IAM users for each identity.
- Create roles in IAM. Manage permissions to control which operations can be performed by the entity or the AWS service that assumes the role. The entity which is allowed to assume the role can also be defined.

Limitations

A few limitations related to entities and objects under IAM are listed below. For a complete list of AWS IAM limitations, visit https://docs.aws.amazon.com/IAM/latest/UserGuide/reference_iam-quotas.html:

- Each entity (user/group/role) created in IAM needs to be unique and cannot be duplicated. The same entity cannot be created twice using all Capital cases, all lower cases, or a combination of both. The created identity may be alpha-numeric, but not all numerical characters
- Password must be at least eight characters long and contain at least 3 of the following:
 - Uppercase letters
 - Lowercase letters
 - Numbers

- Special characters like @, #, \$, %, etc.
- Role names cannot be edited once they are created.
- You can create 500 roles in an AWS account.
- An alias for an AWS account ID should be between 3 and 63 characters.
- A username and role name should not exceed 64 characters.
- A group name should not exceed 128 characters.
- A policy name should not exceed 128 characters.

Creating an IAM Role

An *IAM Role* is an identity created in your AWS account with specific permissions. An IAM role is similar to that of an IAM user regarding permission policies assigned that determine access levels provided. Roles can be used to assign access to users, applications, or services that do not have access to your AWS resources.

IAM roles can be used to delegate access to your AWS resources and create trust relationships between your *trusting account* and other *AWS trusted accounts*. The *trusting account* is the owner of the resource that needs to be accessed, and the *trusted account* contains the users who need to access the resource. For example, allowing a third-party vendor to access your applications for a security audit. In this case, you will be the *trusting account* and the third-party vendor will become the *trusted account*.

IAM roles can be created using any one of the following methods:

- Using Console
- Using AWS CLI (Command Line Interface)
- Using AWS API (Application Programming Interface)
- Using AWS CloudFormation

In this article, the process of creating a role using Console is explained. A detailed explanation of how a role can be created using the other methods can be viewed in the **Creating a role to delegate permissions to an IAM user** section of the **IAM User Guide** section in the **AWS Identity and Access Management Documentation** website. To access these details, [click here](#).

Creating an IAM Role using Console

Prerequisites

- Existing AWS Account
- Existing IAM user with Administrator privileges

Procedure

1. Enter the URL <https://console.aws.amazon.com/iam/> to access the **AWS Management Console**.
2. Select the account type as **IAM user** to open the **AWS Console**.
If there are no IAM users available with Administrator privileges, select the account type as Root user.

*The **Root user** has access to all features of IAM and should only be used with utmost care and only when required.*

3. Select **Roles** → **Create role** in the console's navigation pane.

4. Select the **AWS account** role type.

5. Select **This account** to create a role for your account.

(or)

Select **Another AWS account** and enter the **Account ID**, which needs to access your resources to create a role for another account,

The administrator of the specified account can grant permission to any IAM user in that account to assume the newly created role.

(or)

If there are users from accounts that you do not control that need to assume this role, select **Require external ID**. The external ID can be any word or number agreed on by the administrator of the third-party account and your organization.

*Choosing this option will restrict access to the role only through the **AWS CLI**, **Tools for Windows PowerShell**, or the **AWS API**.*

6. Select **Require MFA** if you want to restrict access to this role only for users who sign in using Multi-Factor Authentication (MFA).

7. Every time a user wanting to assume this role signs in, a temporary one-time password from a configured MFA device will be generated and sent to the user. They can only access this role if they enter this one-time password.

8. Select **Next**.

9. Each role must be aligned with a permission policy to ensure that the created role can only access the resources needed. Select the relevant policy from the list of the AWS-managed and customer-managed policies in your account if they have been created. If no policies have been created, Select **Create policy** to open a new browser tab and create a new policy. To know how to create a new policy, refer to the **Creating an IAM Policy** section of this document.

If you prefer, you can assign a policy to this role at a later time. By default, no permissions are assigned to the created role.

10. To set a permissions boundary,

- Access the **Set permissions boundary** section
- Select **Use a permissions boundary** to control the maximum role permissions.
- Select the policy to use for the permissions boundary.

This is an advanced feature and not mandatory.

11. Select **Next**.

12. Enter the name of the role in the **Role name** field.

Each role name created in your AWS account must be unique and cannot be duplicated.

13. Enter a description of the role in the **Description** field, if required.

14. Edit the use cases and permissions for the role by selecting **Edit** in the **Step 1: Select trusted entities** or **Step 2: Add permissions** sections.

15. Add metadata to the role by attaching tags as key-value pairs by selecting **Next: Tags**. To learn further details on using tags in IAM, refer to the **Tagging IAM resources**

section of the **IAM User Guide** section on the **AWS Identity and Access Management Documentation** website. To access these details, [click here](#).

16. Review the details entered and select **Create Role** to create the role.

Role names cannot be edited once they are created.

Once the role is created, permission needs to be given for users to assume this role or to switch to this role in the console. To know how to perform this action, refer to the **Granting a user permission to switch roles** section of the **IAM User Guide** section on the **AWS Identity and Access Management Documentation** website. To access these details, [click here](#).

Creating an IAM Policy

In IAM, a *policy* is an entity that defines the permissions/actions that users, groups, or roles have access to. The AWS Management Console can create/manage customer-managed policies in IAM. Customer-managed policies refer to policies created/customized and administered in your AWS account. These policies are then assigned to your AWS account's relevant identities (users, groups, and/or roles).

IAM policies can be created using any one of the following methods:

- Using the JSON tab
- Using the visual editor
- Importing existing managed policies

In this article, the process of creating a policy using the JSON tab is explained. A detailed explanation of how a policy can be created using the other methods can be viewed in the **Creating IAM policies** section of the **IAM User Guide** section on the **AWS Identity and Access Management Documentation** website. To access these details, [click here](#).

Prerequisites

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3. Select **Policies** → **Create policy** in the navigation pane of the console.
4. Access the **JSON** tab.
5. Type or paste a JSON policy document. To understand how the policies are created using JSON, refer to the **IAM JSON policy reference** section of the **IAM User Guide** section on the **AWS Identity and Access Management Documentation** website. To access these details, [click here](#).

6. Validate the policy document created to check for security warnings, errors, or general warnings. To understand the scope of policy validation, refer to the **Validating IAM policies** section of the **IAM User Guide** section on the **AWS Identity and Access Management Documentation** website. To access these details, [click here](#).
 7. Resolve any warnings or errors generated during the policy validation.
 8. Select **Review Policy**.
 9. Type a **Name** and **Description** (if needed) for the created policy in the respective fields.
 10. Review the policy summary in the Summary field to see the permissions granted for the created policy.
 11. Select **Next: Tags** to add metadata to the policy by attaching tags as key-value pairs. To learn further details on using tags in IAM, refer to the **Tagging IAM resources** section of the **IAM User Guide** section on the **AWS Identity and Access Management Documentation** website. To access these details, [click here](#).
 12. Select **Create Policy** to add the policy to the list of existing policies.
- Once the policy has been created, it can be assigned to users, groups, and/or roles. To know the process of assigning a policy, refer to the **Adding and removing IAM identity permissions** section of the **IAM User Guide** section in the **AWS Identity and Access Management Documentation** website. To access these details, [click here](#).